Covariance

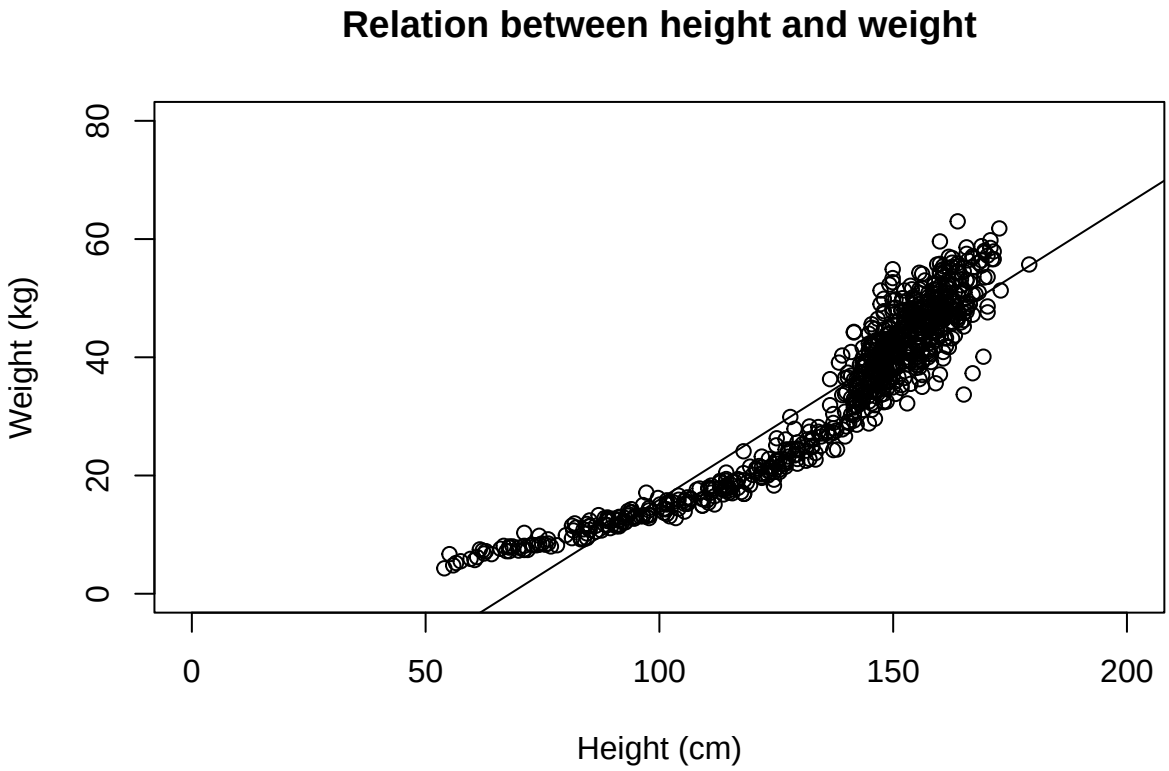
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Covariance

I here try to give a few intuitions about covariance. Overall, it is a measure of how dependent two variables are upon one another; i.e. if you know something about one of them, do you also know something about the other. E.g. with height and weight, knowing one makes you know something about the other. In fact, that might be why we think it is appropriate to model their relationship, e.g. with a linear function.

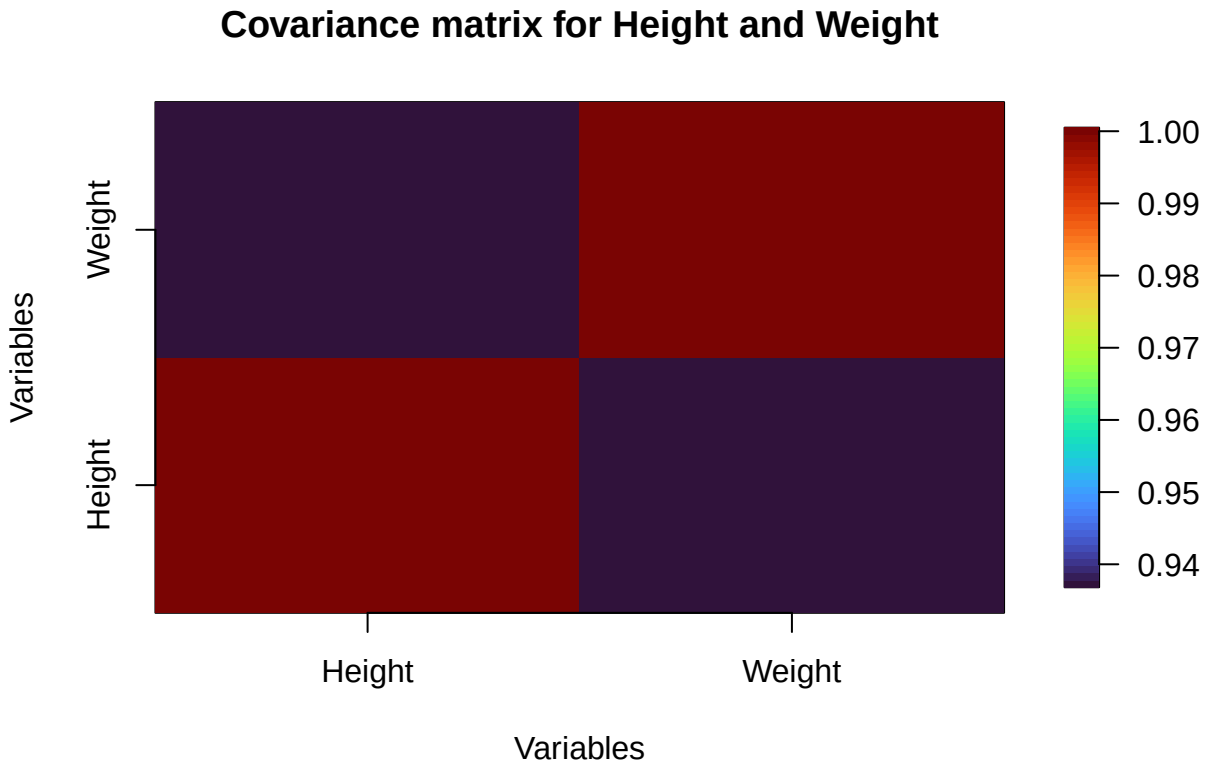
```
library(lgrdata)
data(howell)
plot(weight ~ height, data=howell, xlab='Height (cm)',
     ylab='Weight (kg)',
     main='Relation between height and weight',
     xlim=c(0, 200), ylim=c(0, 80))
model <- lm(weight ~ height, data=howell) ## note we are not capturing the fit perfectly
abline(model)
```



Let's calculate the covariance

First, we put them in one matrix, X, so we can calculate the covariance $X^T X$. We will standardise the data by subtracting the mean and dividing by the standard deviation of each column. This will result in the data being approximately normally distributed with $\mu = 0, \sigma = 1$. This makes sure that the covariance can be calculated as a percentage for ease of comparison, $\frac{1}{n-1} X^T X$

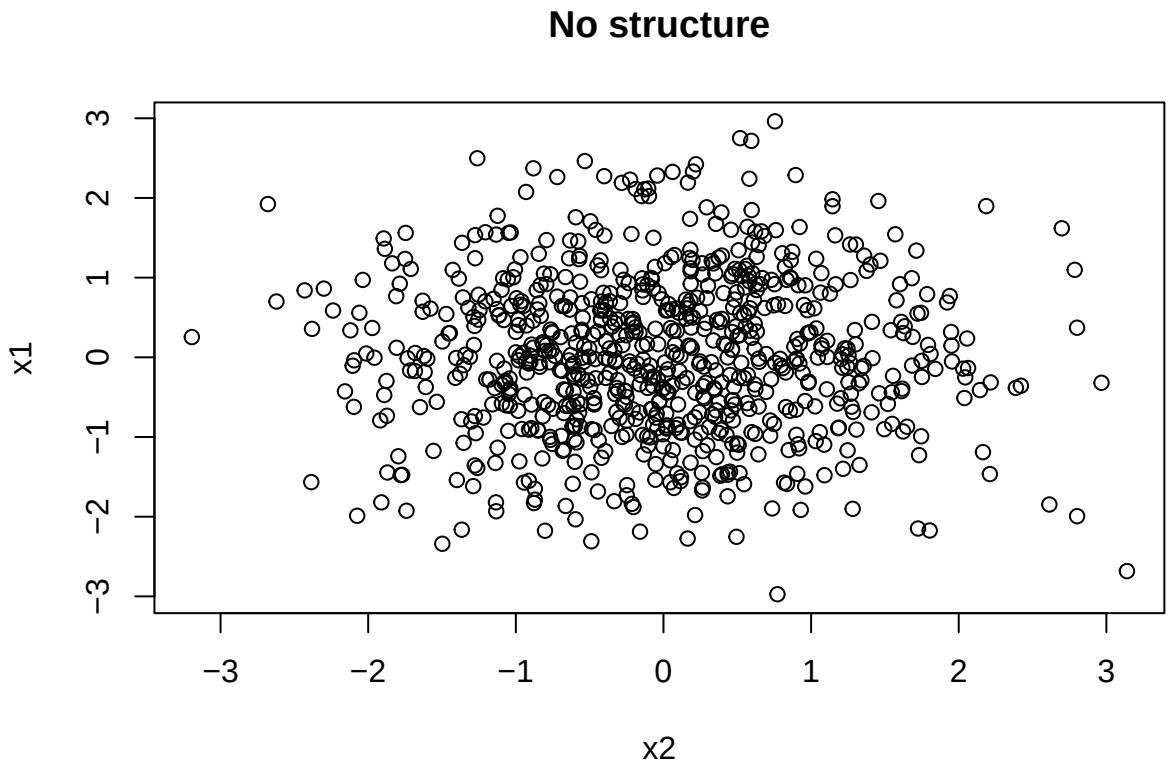
```
suppressMessages(library(fields)) ## image.plot
howell$height.stan <- (howell$height - mean(howell$height)) / sd(howell$height)
howell$weight.stan <- (howell$weight - mean(howell$weight)) / sd(howell$weight)
X <- matrix(c(howell$height.stan, howell$weight.stan), ncol=2)
n <- dim(howell)[1]
cov <- 1/(n-1) * t(X) %*% X
## let's plot it
image.plot(x=1:dim(cov)[1], 1:dim(cov)[1], cov, xlab='Variables',
           ylab='Variables', xaxt='n', yaxt='n', main='Covariance matrix for Height and Weight')
axis(1, 1:dim(cov)[1], 1:dim(cov)[1], labels=c('Height', 'Weight'))
axis(2, 1:dim(cov)[1], 1:dim(cov)[1], labels=c('Height', 'Weight'))
```



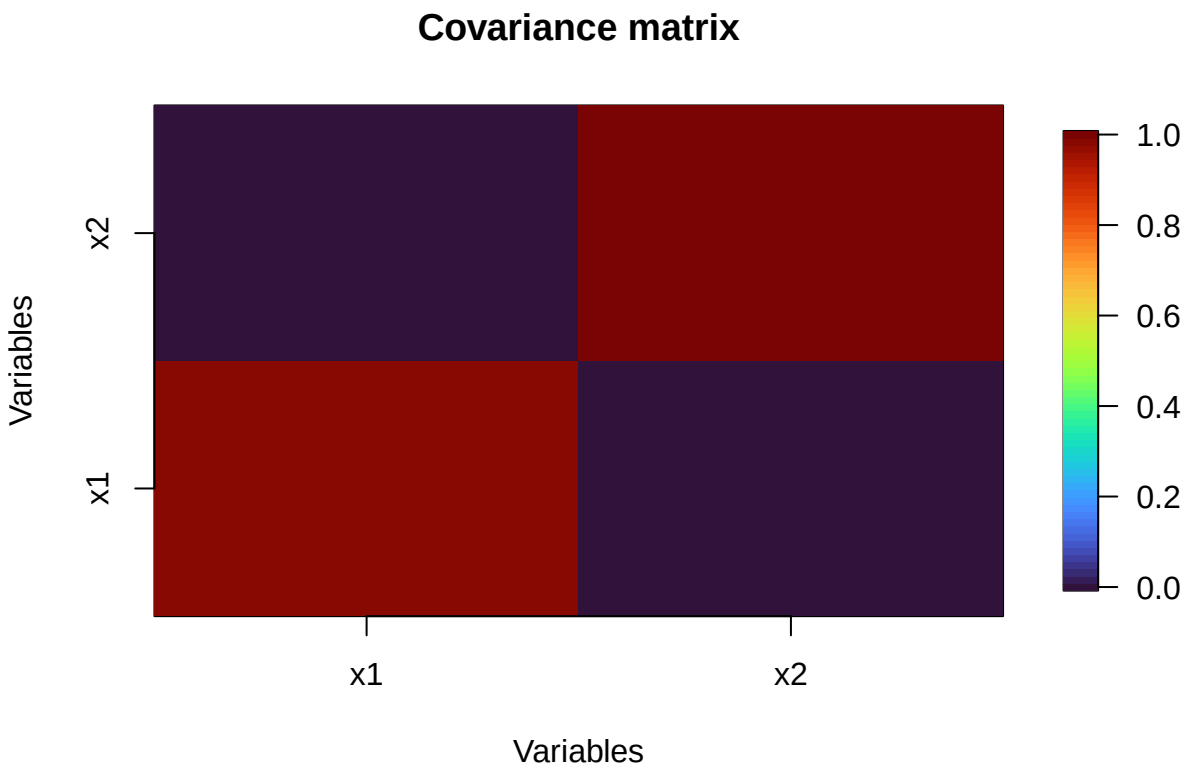
In fact, knowing one variable, means that you "know" 94 % of the other
How does this compare with n observations from two unrelated variables, simulated by drawing from `rnorm` twice?

```
set.seed(7)
x1 <- rnorm(n, mean=0, sd=1)
x2 <- rnorm(n, mean=0, sd=1)

plot(x2, x1, main='No structure')
```




```
X <- matrix(c(x1, x2), ncol=2)
cov <- 1/(n-1) * t(X) %*% X
## let's plot it
image.plot(x=1:dim(cov)[1], 1:dim(cov)[1], cov, xlab='Variables',
           ylab='Variables', xaxt='n', yaxt='n', main='Covariance matrix',
           zlim=c(0, 1))
axis(1, 1:dim(cov)[1], 1:dim(cov)[1], labels=c('x1', 'x2'))
axis(2, 1:dim(cov)[1], 1:dim(cov)[1], labels=c('x1', 'x2'))
```



Note that the covariance between x1 and x2 is close to zero this time.

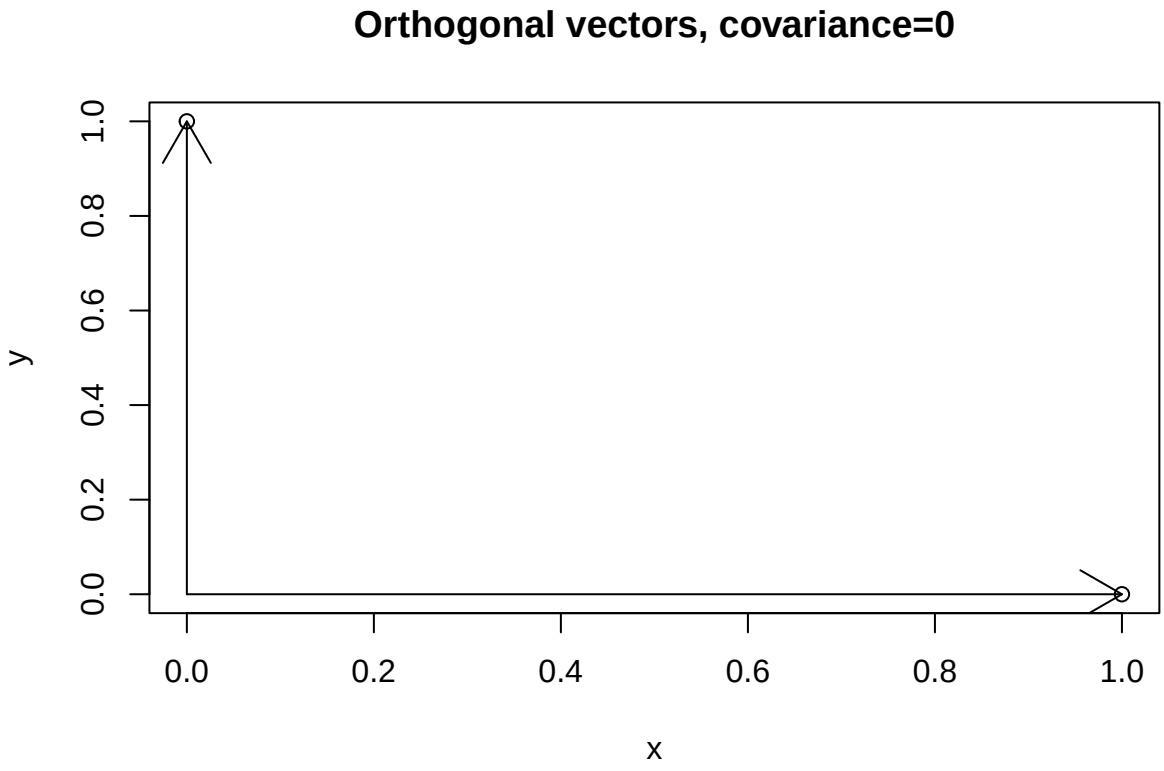
Graphical representation

Now, we will do it with two vectors instead

```
x <- c(1, 0)
y <- c(0, 1)

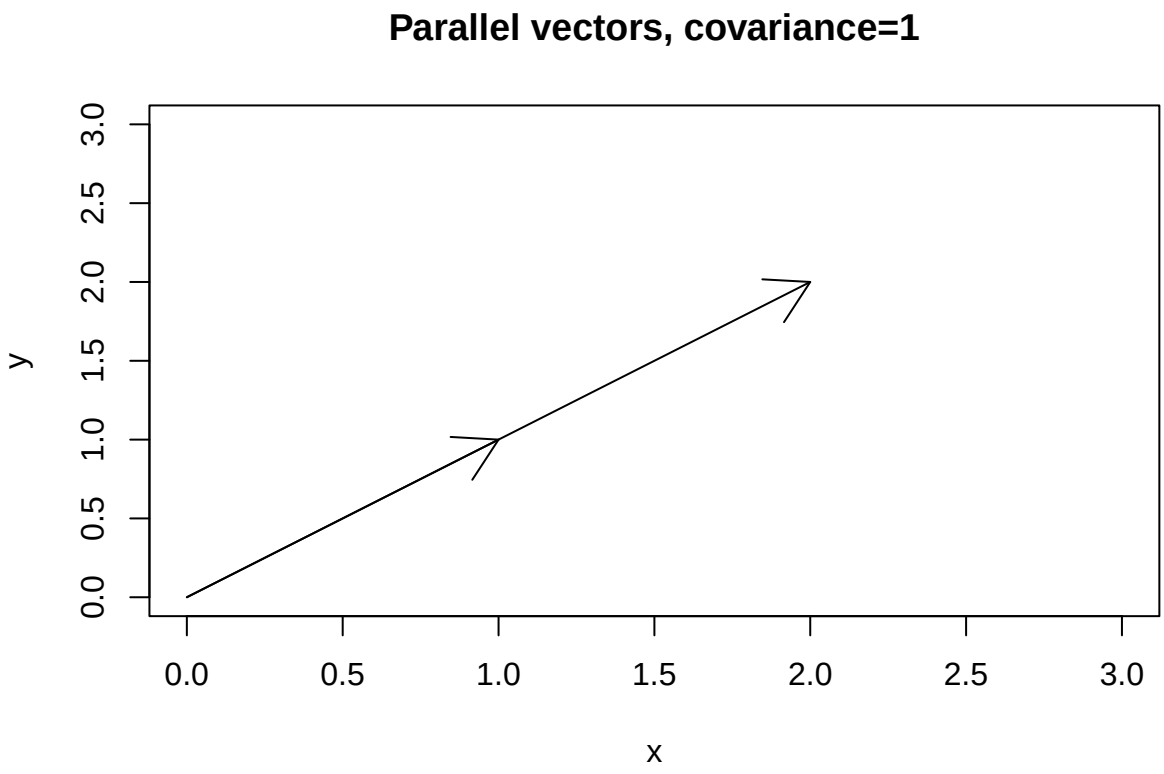
X <- matrix(c(x, y), ncol=2)

plot(x, y, main='Orthogonal vectors, covariance=0')
arrows(0, 0, 1, 0)
arrows(0, 0, 0, 1)
```




```
x <- c(1, 1)
y <- c(2, 2)

plot(x, y, main='Parallel vectors, covariance=1',
     xlim=c(0, 3), ylim=c(0, 3), type='n')
arrows(0, 0, 1, 1)
arrows(0, 0, 2, 2)
```



Matrix multiplication

Let' now try with two matrices,

```
x1 <- c(2, 4, 6)
x2 <- c(6, 9, 3)

X <- matrix(c(x1, x2), ncol=2)

print(t(scale(X)) %*% scale(X))
```

```
##      [,1] [,2]
## [1,]    2   -1
## [2,]   -1    2
```

$$x_1 = \begin{bmatrix} 2 \\ 4 \\ 6 \end{bmatrix}, x_2 = \begin{bmatrix} 6 \\ 9 \\ 3 \end{bmatrix} X = \begin{bmatrix} 2 & 6 \\ 4 & 9 \\ 6 & 3 \end{bmatrix}$$

and now standardised

$$\mu_{x_1} = 4, \mu_{x_2} = 6 \sigma_{x_1} = 2, \sigma_{x_2} = 8$$

$$X_{stan} = \begin{bmatrix} -1 & 0 \\ 0 & 1 \\ 1 & -1 \end{bmatrix}$$

$$X_{stan}^T X_{stan} = \begin{bmatrix} -1 & 0 & 1 \\ 0 & 1 & -1 \end{bmatrix} \begin{bmatrix} -1 & 0 \\ 0 & 1 \\ 1 & -1 \end{bmatrix}$$

$$X_{stan}^T X_{stan} = \begin{bmatrix} -1 \times -1 + 0 \times 0 + 1 \times 1 & -1 \times 0 + 0 \times 1 + 1 * -1 \\ 0 \times -1 + 1 \times 0 + -1 \times 1 & 0 \times 0 + 1 \times 1 + -1 \times -1 \end{bmatrix}$$

$$X_{stan}^T X_{stan} = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$$

$$\frac{1}{n-1} X_{stan}^T X_{stan} = \begin{bmatrix} 1 & -0.5 \\ -0.5 & 1 \end{bmatrix}$$

Covariance between x_1 and $x_2 = -0.5$